Behaviour Cloning via Supervised Learning of an Arm Manipulator on a Moving ase

Noah Calati
Computer Science Dept
UCL
noah.calati.26@ucl.ac.uk

Sam Reed
Computer Science Dept
UCL
samuel.reed.26@ucl.ac.uk

Finn Gorman
Computer Science Dept
UCL
finn.gorman.26@ucl.ac.uk

***Abstract*—TODO: explains (1) the task : what are we doing: (2) how expert data was generated; (3) our model choices (regressor vs decision tree) (4) how we evaluated generalisation, robustness and efficiency; (5) mean final end-effector error and success rate on unseen targets.**

I. Introduction

TODO: Explain behaviour cloning (BC) + learning a policy from expert state + action pairs by supervised regression + [cite] Note its known weakness [cite], evaluation linked to this.

A. Problem Formulation

TODO: Explain that at each control step what measurements are collected e.g.

$$\mathbf{o}_t = \left[\, \mathbf{p}_t^b,\ \psi_t,\ \mathbf{q}_t,\ \dot{\mathbf{q}}_t,\ \mathbf{x}^* \,\right], \tag{1}$$

where $\mathbf{p}_t^b$ and ψ_t are the base position and bearing (`base_pos`, `base_bearing`), $\mathbf{q}_t$ and $\dot{\mathbf{q}}_t$ the measured arm joint positions and velocities (`q_mes`, `qd_mes`), and $\mathbf{x}^* \in \mathbb{R}^3$ the target end-effector position (`TARGET_EE_POS`). It predicts

$$\hat{\mathbf{a}}_t = \pi_\theta(\mathbf{o}_t) = \left[\, \hat{\boldsymbol{\tau}}_t,\ \hat{\mathbf{u}}_t \,\right], \tag{2}$$

the arm torques (`torque_arm`) and wheel commands (`wheel_cmds`).

TODO: state the dimensions (number of arm joints, number of wheels).

B. Assumptions and Constraints

TODO: explain our assumptions and so on (examples listed below)

- obstacles, flat ground e.t.c
- target within bounds or out of bounds...
- Fixed control period (state value).

C. Contributions

[TODO : methodology] TODO: 2–3 bullet points of what you actually did, e.g. an input representation using the target relative to the base; a comparison of single-instant vs. temporal-window inputs; a closed-loop evaluation under noise and unseen initial conditions.

TABLE I
Expert Dataset Summary

Property	Value
Episodes (train / val / test)	TODO / TODO / TODO
Samples (state–action pairs)	TODO
Control frequency (Hz)	TODO
Target range x, y, z (m)	TODO
Initial base pose range	TODO
Initial joint range (rad)	TODO
Expert success rate (%)	TODO

II. Methodology

A. Simulation and Expert Controller

TODO: Describe the simulator and how the provided expert computes arm torques and wheel commands (e.g. task-space/inverse-kinematics control plus a PD law + citations. Give the control frequency, parameters and so on and episode termination condition. Explain what was modified added. (exerts)

B. Expert Data Generation

TODO: Number of episodes; how targets and initial conditions (base pose, joint configuration) were sampled; episode length; logging rate; how you validated data quality (e.g. discarded episodes where the expert failed). Split train/validation/test *by episode*, not by time step, to avoid leakage between correlated consecutive samples. Summarize in Table I.

C. Input and Output Representation

TODO: Justify the final input vector. Discuss (and later test in Table) IV):

- *Extra inputs*: e.g. target expressed relative to the base frame, current end-effector position, or end-effector-to-target error, which make the mapping easier to learn.
- *Single instant vs. temporal window*: stacking the last k observations, or a recurrent model + citation? .
- *Normalisation*: per-feature standardisation of inputs and outputs using training-set statistics only; wrapping of angles (e.g. bearing as $\sin\psi, \cos\psi$).

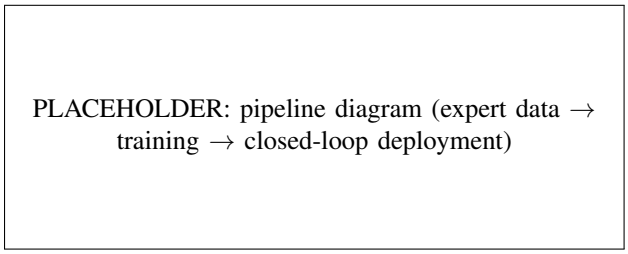


Fig. 1. Overview of the behaviour cloning pipeline.

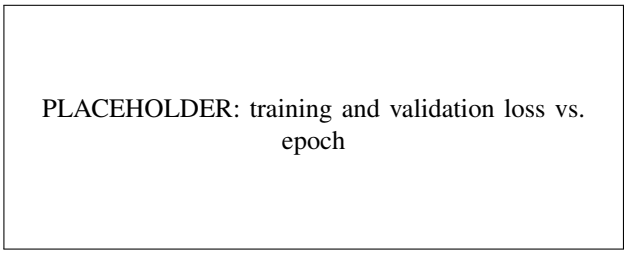


Fig. 2. Training and validation loss for each model.

TABLE II
TRAINING HYPERPARAMETERS

Hyperparameter	Value
Hidden layers × width	TODO
Activation	TODO
Optimiser / learning rate	TODO
Batch size	TODO
Epochs / early stopping	TODO
Weight decay / dropout	TODO
History length k	TODO

D. Model Architecture

TODO: Describe each model you compare.

1) Linear regression
2) tree regression...
3) classification?
4) can include: linear regression, logistic, decision tree, random forest, knn, gradient boosting... (unsure)

Report parameter counts in Table III. CONSIDER a figure of the full pipeline (Fig. 1).

E. Training Procedure

Parameters are fitted by minimising the mean squared error over the training set $\mathcal{D}$:

$$\mathcal{L}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{o},\mathbf{a})\in\mathcal{D}} \left\|\pi_\theta(\mathbf{o}) - \mathbf{a}\right\|_2^2. \tag{3}$$

TODO: Optimiser, framework + citations and how hyperparameters were chosen (validation set, grid search). List final values in Table II. If you add noise to the expert's actions or states during data collection to make the policy more robust + citation.

F. Evaluation Protocol

TODO: Define your metrics:

- **Accuracy**: (i) offline test MSE on held-out episodes; (ii) closed-loop final end-effector error $e = \|\mathbf{x}_T^{ee} - \mathbf{x}^*\|_2$ and success rate ($e <$ TODO cm) when the learned policy replaces the expert.
- **Generalisation**: unseen targets inside the training range (interpolation) and outside it (extrapolation); unseen initial base poses and joint configurations.
- **Robustness**: Gaussian noise on observations, external pushes / perturbations, actuator noise.

TABLE III
CLOSED-LOOP PERFORMANCE ON HELD-OUT TARGETS (MEAN ± STD OVER TODO EPISODES)

Model	Test MSE	EE err (cm)	Succ. (%)	Params	Inf. (ms)
Expert	–	TODO	TODO	–	TODO
Linear	TODO	TODO	TODO	TODO	TODO
MLP	TODO	TODO	TODO	TODO	TODO
Temporal	TODO	TODO	TODO	TODO	TODO

- **Efficiency**: number of parameters, memory, training time, per-step inference time vs. the control period.

Use a fixed set of TODO test targets and random seeds for every model so comparisons are fair; report mean ± standard deviation.

III. RESULTS

A. Training and Offline Accuracy

TODO: Training/validation loss curves (Fig. 2); comment on over/underfitting. [potentially] - might not be linked to what we have learnt

B. Closed-Loop Accuracy and Efficiency

TODO: Main comparison (Table III) and an example rollout, expert vs. learned (Fig. 3). Interpret.

C. Ablation Studies

TODO: Effect of input choices, temporal window and data quantity (Table IV; a plot of error vs. number of training episodes also works well here).

D. Generalisation and Robustness

TODO: Results in Table V. Relate failures to the training distribution and to compounding error [3].

IV. CONCLUSION

TODO: 2/3 sentence summary of findings with numbers.

Limitations. TODO: e.g. compounding error / covariate shift, dependence on the training distribution, simulation only, limited number of test targets.

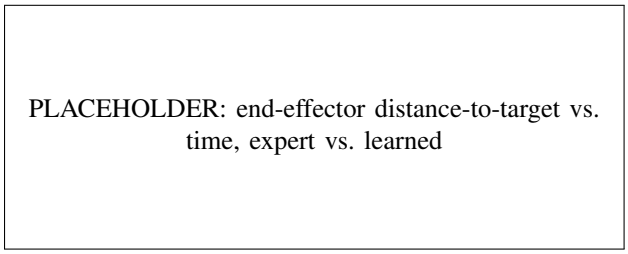


Fig. 3. Expert and learned rollouts for an unseen target.

TABLE IV
ABLATION OF INPUTS, HISTORY LENGTH AND DATA QUANTITY

Variant	**Test MSE**	**EE err (cm)**
Raw inputs (1)	TODO	TODO
+ relative target / EE error	TODO	TODO
+ history $k =$ TODO	TODO	TODO
TODO % of training episodes	TODO	TODO
TODO % of training episodes	TODO	TODO
100 % of training episodes	TODO	TODO

REFERENCES

[1] D. A. Pomerleau, "ALVINN: An autonomous land vehicle in a neural network," in *Advances in Neural Information Processing Systems*, 1989, pp. 305–313.

[2] B. D. Argall, S. Chernova, M. Veloso, and B. Browning, "A survey of robot learning from demonstration," *Robotics and Autonomous Systems*, vol. 57, no. 5, pp. 469–483, 2009.

[3] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. Int. Conf. Artificial Intelligence and Statistics (AISTATS)*, 2011, pp. 627–635.

[4] F. Codevilla, E. Santana, A. M. López, and A. Gaidon, "Exploring the limitations of behavior cloning for autonomous driving," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 9329–9338.

[5] M. Laskey, J. Lee, R. Fox, A. Dragan, and K. Goldberg, "DART: Noise injection for robust imitation learning," in *Proc. Conf. Robot Learning (CoRL)*, 2017, pp. 143–156.

[6] T. Z. Zhao, V. Kumar, S. Levine, and C. Finn, "Learning fine-grained bimanual manipulation with low-cost hardware," in *Proc. Robotics: Science and Systems (RSS)*, 2023.

[7] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[8] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2015.

[9] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, 2019, pp. 8024–8035.

[10] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, *Robotics: Modelling, Planning and Control*. London, U.K.: Springer, 2009.

[11] E. Coumans and Y. Bai, "PyBullet, a Python module for physics simulation for games, robotics and machine learning," 2016–2021. [Online]. Available: http://pybullet.org

TABLE V
GENERALISATION AND ROBUSTNESS OF THE BEST MODEL

Test condition	**EE err (cm)**	**Succ. (%)**
In-distribution targets	TODO	TODO
Out-of-distribution targets	TODO	TODO
Unseen initial base pose	TODO	TODO
Unseen initial joint configuration	TODO	TODO
Observation noise $\sigma =$ TODO	TODO	TODO
Observation noise $\sigma =$ TODO	TODO	TODO
External perturbation	TODO	TODO

Contribution Statement

- **Noah Calati**: TODO (e.g. data generation pipeline, Sections II-B–II-C).
- **Sam Reed**: TODO (e.g. model implementation and training, Sections II-D–II-E).
- **Finn Gorman**: TODO (e.g. evaluation experiments and results, Section III).

All members contributed to writing and reviewing the report.